Stats 140XP Team 6 Final Project

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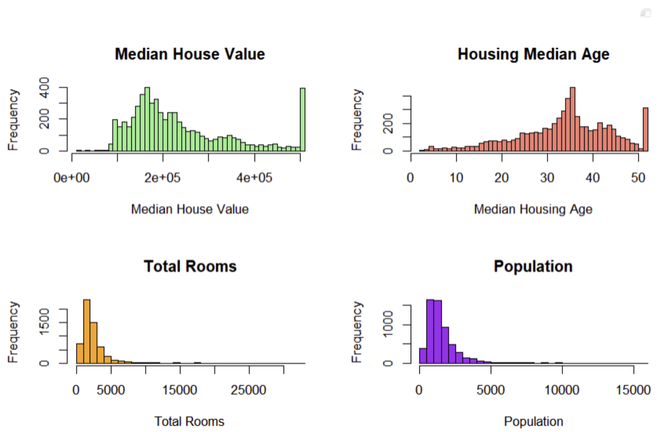
Introduction

Housing prices in Los Angeles have risen dramatically over the past decades, driven by various social and economic factors. Understanding which factors strongly influence housing prices is important for consumers and policymakers. With increasing availability of housing data and modern statistical methods, this project will explore these relationships and build predictive models of housing prices.

- **Our research objectives are to:**
  (1) analyze whether structural housing characteristics such as total rooms and housing age contribute to property price variation;
  (2) examine whether population and household density influence pricing patterns;
  (3) explore engineered features such as distance to landmarks (e.g., nearest school, Hollywood) to improve prediction accuracy.
- **Data Source:** Our chosen datasets: **(1) 1990 California Housing Price**, and **(2) 2024 California Housing Listing (Zillow)** originates from Kaggle. We filtered the data into LA county (1990) and LA city (2024). Since the second dataset does not have coordinates, we masked the first dataset coordinates onto the second one.

AI Statement: Some stylistic formatting by the LaTeX source code, and sentence polishing is assisted by ChatGPT GO.

Exploratory Data Analysis



These four plots shows several important features of LA housing dataset. The median house value distribution demonstrates a right-skewed with a noticeable spike on the right, indicating the cap of housing price. The housing median age is left-skew and also has a ceiling. Total rooms and population exhibits strong right-skewed pattern, implying the presence of few high population density areas.

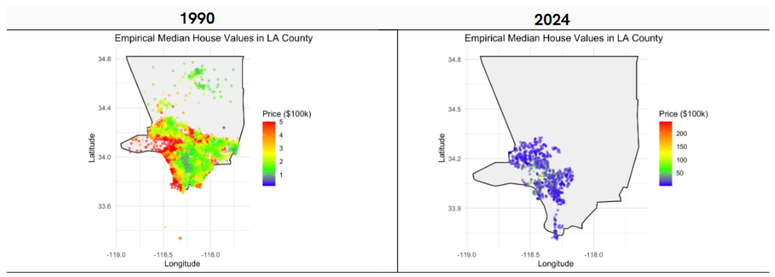


Figure 1. Empirical Median House Values in LA County in 1990 Figure 2. Empirical Median House Values in LA County in 2024

Methodology

Statistical Methods
Our strategy is to run simple to complex models, utilizing techniques like feature importance and regularization to improve findings. Then we conduct cross validation and benchmark experiments to test model performance. Additionally, we explored additional features Kriging interpolation for our best model - Random Forest on log-transformed data. This feature enables a smooth heatmap prediction in LA city by algorithmically interpolating empty space between neighboring values.

Testable Hypotheses

Geographic Effects
H0: Geographic location is not significantly associated with property value.
HA: Geographic location significantly predicts property value, indicating a spatial price gradient across Los Angeles.

Structural Characteristics
H0: Structural housing characteristics do not significantly predict property value.
HA: Structural housing characteristics significantly predict property value.

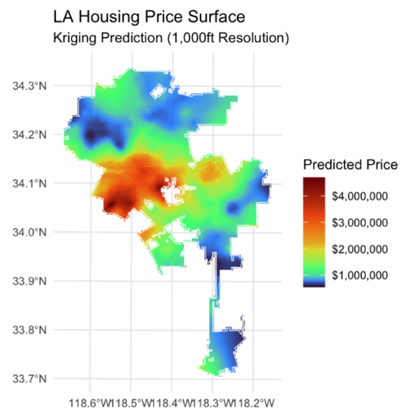
Benchmark Experiment

Method Name	Inputs	RMSE	R^2
Multilinear Regression	Raw Price	960,062	0.7741
Ridge	Base + regularization	880,954	0.8098
Lasso	Raw Price	882,738	0.8090
Random Forest	Nonlinear interactions	1,353,676	0.5509

Since most of the features are heavily skewed to the right (i.e. large clusters on the left hand side). Log transformation is a good tool to reduce skewness and make the transformed distribution approximating normality. Also, log-transformation significantly increases the percentage of variance explained (R^2) by Random Forest.

Method Name	Inputs	RMSE	R^2
Multilinear Regression	Log(Price)	0.7417	NA
Ridge	Base + regularization	0.4916	0.4996
Lasso	Log Transformation	0.6485	NA
Random Forest	Nonlinear interactions	0.3264	0.7794

Prediction



This map shows a predicted housing price surface across Los Angeles using kriging interpolation. This method estimates prices for locations without observations by weighting nearby property values. Warmer colors indicate higher predicted prices, while cooler colors indicate lower prices. The visualization highlights geographic clustering of housing values across the city.

Data Processing & Feature Engineering

Drop Variables: Drop unneeded or redundant variables such as `zpid`, `streetAddress`, `dateSold`.

City Filtration: Filtering the Data limited to LA County.

One-Hot Encoding: Dummy coding categorical variables to examine the effects in each subgroup.

Transform Variables: Replacing `yearBuilt` with `house_age = 2024 - yearBuilt`, and transforming `datePosted` data type using `as.Date()`.

Setting Boundary: Filtering `price` above 200k.

Data Transformation: log-transforming heavily left-skewed variables such as `totalRooms` and `population`.

Conclusion

Our analysis shows that several structural housing characteristics play an important role in explaining variation in Los Angeles housing prices. Latitude also showed statistical significance, suggesting the presence of spatial price patterns across Los Angeles, while other variables contributed less strongly to the model.

- Living Area: Larger homes strongly associated with higher prices
- Rent Zestimate: Strong positive correlation with property value
- Latitude: Indicates spatial price gradient across Los Angeles
- Several structural variables showed weak significance in the baseline model

Modeling Results
To improve predictive performance, we compared several modeling approaches, including **Multiple Linear Regression, Ridge Regression, Lasso Regression, and Random Forest.** Because housing prices are heavily right-skewed, applying a **log transformation** significantly improved model fit, allowing the models to capture relationships in the data better.
Among the models tested, Random Forest applied to log(price) produced the strongest overall performance, achieving the lowest prediction error on the transformed scale and capturing nonlinear relationships between housing features.

- Baseline MLR achieved RMSE $\approx$ \$960k
- Regularized models (Ridge/Lasso) improved predictive accuracy
- Random Forest with log(price) produced the best overall performance
- Results suggest nonlinear relationships remain important in housing price prediction

Future Work

- Baseline multilinear regression achieved RMSE of approximately \$794k
- Structural and geographric variables explain some price variation
- However, large prediction error suggests nonlinear relationships remain

References (APA 7)

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